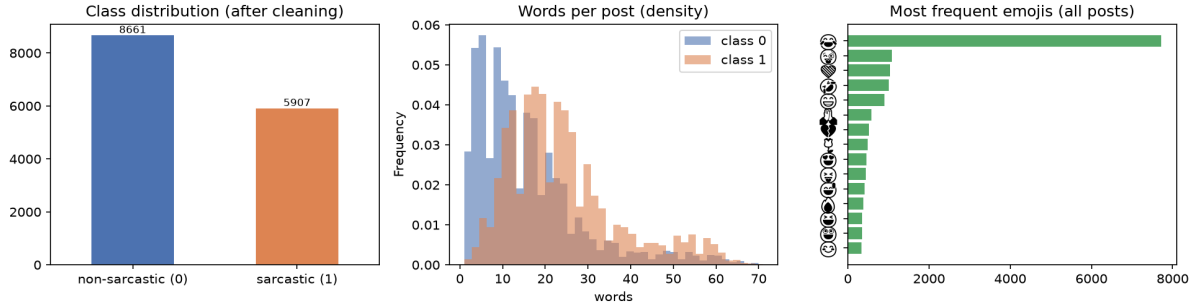
Urdu Sarcasm Detection with Emoji Semantics

DS5007 Natural Language Processing (Fall 2026) — Programming Assignment 1 — Kiran (kiran.poulta@gmail.com). Code, notebook and outputs: notebooks/Urdu_Sarcasm_Detection.ipynb, src/, outputs/.

1. Data and preparation

The provided sheet has 20,004 rows. Cleaning removed 49 rows without text, 4,095 exact duplicate posts (which would otherwise leak between train and test) and 646 texts that occur with **both** labels (1,292 rows; unlearnable annotation conflicts). 14,568 posts remain: 8,661 non-sarcastic / 5,907 sarcastic (59/41 %) — the raw file only looked balanced because the sarcastic class was heavily duplicated. A stratified 70/15/15 split gives 10,196/2,186/2,186 posts; all model selection uses the validation split, the test split is used once.

Posts are short (median 16 words, max 71); sarcastic posts are longer (median 20 vs 13 words). 79% of posts contain emojis (88 % of non-sarcastic vs 67 % of sarcastic — the non-sarcastic class is dominated by poetry and greetings decorated with 💚 🌹 💔). 15% contain Latin-script tokens (code-mixing: ISPR, PM, hashtags), 7% hashtags (often scraped as words_with_underscores), 5% mentions, 3% URLs, 2% quoted speech (`` … ''). Further quality issues: Arabic/Urdu code-point variants (ك/ک, ي/ی), ھ typed for ہ (ھے), elongation (کدووووو), ~8 % Punjabi in Urdu script, and collection artefacts — some posts still carry the harvesting hashtags #طنز / #مزاح / #sarcasm, which a linear model learns as its strongest feature. Many clearly mocking political posts are labelled 0 and some plain shayari with 💔 is labelled 1, which caps the attainable accuracy for every model.



2. Urdu-aware preprocessing and emoji semantics

Normalisation (src/preprocessing.py) is conservative and every step is switchable for ablation: Unicode NFC and code-point unification (ك→ک, ي→ی, ة→ہ, أ→ا, Sindhi ئ→ئ, Urdu digits→ASCII); diacritic removal; whole-word keyboard fixes (ہے→ھے, ہیں→ھیں, کوئی→کوئ — ھ is a real letter in تھا so only listed words are changed); URL→URL, @user→@USER; hashtag reconstruction (#دو_عزت_کو_ووٹ → HASHTAG دو عزت کو ووٹ); laughter (ہاہاہا, hahaha, lol→LAUGH) and elongation (کدووووو→کدوو) normalisation; Urdu punctuation mapped to ASCII but **kept** as tokens because rhetorical ؟؟؟ and … are cues. Roman/English and Punjabi tokens are kept. Tokenisation is regex-based (Urdu word runs, Latin words, numbers, punctuation runs, single emojis). Emojis are extracted before normalisation because the zero-width joiner we strip from Urdu text is part of emoji sequences (👰‍♂️).

Emoji semantics (src/emoji_semantics.py). A curated lexicon of ~150 emojis gives each an English name, an **Urdu gloss** (😂 آنسو, 😊 آنکھیں گھمانا, 💔 دل ٹوٹا), a polarity (+/0/−), a pragmatic category (LAUGH, TEASE, SMIRK, THINK, LOVE, SAD, ANGER, PRAYER, PRAISE, FLOWER, SHY, GESTURE, INSULT) and an irony-cue flag (😊 😌 🤪 😅 🤭 …). Skin tones and variation selectors are canonicalised (💚 = 💚, ✌🏽 = ✌). The lexicon drives: (i) seven representations tested in the ablation — remove, keep raw, meaning token (EMO_face_with_tears_of_joy), category token (EMOCAT_LAUGH), meaning+category+polarity+cue, Urdu gloss (😂 → آنسو), and **augment** (😂 → 😂 آنسو: the symbol as used plus its meaning in the language of the post — the default input of the transformer); (ii) 27 dense emoji features: counts per category, polarity mix, position (start/middle/end), longest repeat run (😂😂😂😂), laughter-only; (iii) a data-driven emoji sarcasm prior (log-odds per emoji, fitted on the training split only); (iv) **word–emoji polarity contrast**: positive words or ironic-praise markers (واہ, ماشاءاللہ, کیا بات ہے, شاباش, تبدیلی, خان صاحب) combined with an irony-cue/negative emoji — the canonical sarcasm pattern; (v) an **emoji meaning dictionary** of the 538 emojis in the corpus (96 % of occurrences have a curated Urdu meaning) with P(sarcastic|emoji); and (vi) a **predict-and-explain API** (src/predict.py) that returns, with every label, the meaning of each emoji in English and Urdu, its polarity/category, the cues that fired (ironic praise, idiom, polarity contrast, negation, rhetorical punctuation, repetition, code-mixing) and a one-line reading, e.g. for صاحب کیا بات ہے تبدیلی آ گئی واہ خان 😂😂😊 → sarcastic (0.73); 😂 = بنسی کے آنسو, 😊 = آنکھیں گھمانا; cues: ironic-praise marker, irony-cue emoji, positive words + ironic emoji; reading: the praise words are a vehicle for mockery.

Hand-crafted Urdu phrase features (bonus, src/features.py, 72 features): ironic-praise markers, 35 idioms/multi-word expressions (اپنے پیر پر کلہاڑی, چراغ تلے اندھیرا, دودھ کا راکھا بلا, ٹوپی ڈرامہ, کر لو گل …), negation words, small positive/negative lexicons, question/ellipsis/repetition counts, quoted speech, code-mixing ratio, laughter, hashtags/mentions/URLs. In the training split P(sarcastic|emoji) ranges from 0.00–0.06 for 🙂 🥀 🌹 💕 💚 🌹 💔 to 0.55–0.61 for 😅 😂 😄 😊 🤭 😅 😂 (base rate 0.41); 😂 alone is only mildly informative (0.50) — it marks humour, and the words decide whether the laughter is at someone.

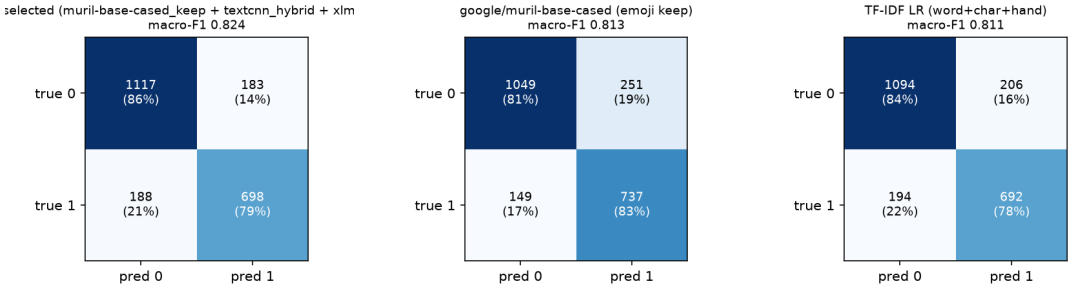
3. Models and experiments

Classical: TF-IDF word uni/bi-grams + character 2–5-grams (within word boundaries; robust to spelling variation and suffix morphology) + the standardised hand-crafted block, with Logistic Regression (liblinear, class-balanced), a calibrated Linear SVM and Complement NB; C tuned on validation. **Neural (PyTorch):** TextCNN (200-d embeddings learnt from scratch, filter widths 2–5 × 128, max-pooling) and BiLSTM (128/direction) + additive attention + max pooling; hybrid variants concatenate the hand-crafted vector; Adam, class-weighted CE, dropout 0.5, early stopping on validation Macro-F1, threshold tuned on validation; every emoji is one token so emoji sequences are visible. **Transformer:** xlm-roberta-base (pre-trained on 100 languages incl. Urdu) fine-tuned with the 250k×768 word-embedding matrix frozen (192M of 278M parameters — fits a 6 GB GPU and regularises on 10k examples; all 12 encoder layers and the head are trained; AdamW 2e-5, warm-up, class-weighted loss, fp16, max 96 sub-words, 3 epochs, early stopping, threshold tuned on validation), with three emoji inputs: symbol + Urdu meaning (augment), raw symbol, Urdu meaning only. Macro-F1 is the primary comparison metric.

4. Results (test split, n = 2 186)

model	acc	P(0)	R(0)	F1(0)	P(1)	R(1)	F1(1)	Macro-F1
Ensemble: validation-weighted (NNLS)	0.833	0.862	0.856	0.859	0.791	0.799	0.795	0.827
Ensemble: greedy-selected (muril-base-cased_keep + textcnn_hybrid + xlm-roberta-base_augment)	0.83	0.856	0.859	0.858	0.792	0.788	0.79	0.824

model	acc	P(0)	R(0)	F1(0)	P(1)	R(1)	F1(1)	Macro-F1
Ensemble: deployable (muril-base-cased_keep + tfidf_lr + textcnn_hybrid)	0.829	0.864	0.845	0.855	0.78	0.805	0.792	0.823
google/muril-base-cased (emoji keep)	0.817	0.876	0.807	0.84	0.746	0.832	0.787	0.813
xlm-roberta-base (emoji keep)	0.814	0.883	0.792	0.835	0.735	0.847	0.787	0.811
TF-IDF LR (word+char+hand)	0.817	0.849	0.842	0.845	0.771	0.781	0.776	0.811
xlm-roberta-base (emoji urdu)	0.817	0.84	0.855	0.847	0.781	0.762	0.771	0.809
xlm-roberta-base (emoji augment)	0.811	0.86	0.815	0.837	0.748	0.806	0.776	0.806
TF-IDF SVM (word+char+hand)	0.815	0.83	0.866	0.848	0.79	0.739	0.764	0.806
TF-IDF LR (word+char, emoji semantics)	0.812	0.851	0.828	0.839	0.758	0.787	0.772	0.806
TF-IDF LR (word+char, emojis removed)	0.81	0.853	0.823	0.838	0.753	0.791	0.772	0.805
TextCNN + hand-crafted (emoji keep)	0.806	0.838	0.835	0.837	0.759	0.764	0.762	0.799
TextCNN (emoji keep)	0.803	0.843	0.823	0.833	0.749	0.774	0.761	0.797
BiLSTM+Attn + hand-crafted (emoji keep)	0.803	0.834	0.834	0.834	0.756	0.757	0.757	0.796
BiLSTM+Attn (emoji keep)	0.79	0.837	0.804	0.82	0.728	0.771	0.749	0.785



Confusion matrices (rows = true, cols = predicted): deployed ensemble, best single transformer (MuRIL), best classical. All families mis-classify more sarcastic posts than non-sarcastic ones (lower recall on class 1), i.e. subtle mockery without lexical markers is the hard case.

Ensemble and model selection. The probability outputs of the transformers, the TF-IDF LR and the hybrid TextCNN are combined; member weights (non-negative least squares) or the member subset (greedy forward selection) and the decision threshold are fitted on the validation split only and applied once to the test split. Combining three inductive biases - sub-word encoder, character n-grams, hand-crafted emoji/phrase cues - is worth about +1.5 Macro-F1 over the best single model, the largest gain of any change in this project, because the members make partly different errors on the noisy posts. The system selected by validation Macro-F1 is **greedy-selected (muril-base-cased_keep + textcnn_hybrid + xlm-roberta-base_augment)** (validation 0.837; test Macro-F1 0.824, accuracy 83.0%); it is the model packaged in outputs/best_model.json and served by app.py.

5. Ablation study (5-fold CV on train+val, TF-IDF LR, mean ± std Macro-F1)

configuration	Macro-F1	± std	accuracy
emojis removed	0.819	0.0071	0.8251
emojis kept raw	0.8222	0.0068	0.8277
emojis -> meaning token	0.8209	0.0058	0.8264
emojis -> Urdu gloss	0.8203	0.0067	0.8259
emojis -> meaning+cat+polarity	0.8208	0.0054	0.8264
emoji symbol + Urdu meaning (augment)	0.8204	0.0081	0.8259
word only	0.8094	0.0055	0.8144
char only	0.8163	0.0024	0.8215
hand-crafted only	0.7559	0.0102	0.7593
word+char+hand	0.8217	0.0063	0.8272
raw text, no normalisation	0.8214	0.0037	0.8269
Urdu char normalisation only	0.8229	0.0041	0.8283
Urdu + social normalisation	0.8222	0.0068	0.8277
artefact hashtags removed (word+char+hand)	0.8153	0.007	0.8208

Reading the ablation (differences below ~0.6 pt are within one fold-std). Emojis: removing emojis costs 0.3 pt for the n-gram model; every meaning-aware representation (meaning token, Urdu gloss, meaning+category+polarity, symbol+meaning) is within noise of the raw symbol. The reason is corpus-specific: once words such as a politician's name, طنز / مزاح, or ironic praise are present the emoji is redundant for a bag-of-words model; the emoji decides the short, lexically neutral posts (see error analysis), and the neural models, which see emoji sequences, gain ~1 pt from keeping them (TextCNN 0.797 vs 0.788). The meaning representations matter for a different reason — rare emojis: 538 distinct emojis occur, 60 % of them fewer than five times, and mapping them to a category / polarity / Urdu meaning lets the model generalise from 😂 to 😒, which the raw symbol cannot. Features: char n-grams beat word n-grams (spelling variation, suffix morphology, no reliable spacing) and the two are complementary (+0.6 pt); the 72 hand-crafted features alone reach 0.756 and add a small consistent gain on top. Normalisation: Urdu character normalisation is the best single setting (+0.15 pt over raw, lowest variance); the social-media steps (laughter / punctuation spacing) are neutral, so rhetorical punctuation is kept. Artefacts: 842 of 12 382 train+val posts still contain the hashtags used to harvest the sarcastic class (#طنز, #مزاح, #sarcasm) and 835 of them are labelled sarcastic; removing these tokens lowers Macro-F1 by 0.6 pt, i.e. all reported scores are slightly optimistic relative to in-the-wild performance and the honest number is ≈ 0.815. Transformer emoji input: raw emoji 0.811, Urdu meaning only 0.809, symbol + meaning 0.806 Macro-F1 on test — also within noise: XLM-R's SentencePiece vocabulary already carries usable representations for the frequent emojis, and the Urdu gloss mainly helps the ~40 % of emoji types that are rare. The practical value of the meaning layer is therefore less in the headline score than in (a) robustness to rare emojis, (b) the dense contrast features for the classical/neural models, and (c) explainability: every prediction can be returned with the meaning of its emojis and the cues that fired.

6. Error analysis

The 25 most confident test-set errors of XLM-R (augment) and of TF-IDF LR were inspected; the two models share 288 of their ~400 errors, so the hard cases are properties of the data rather than of a model family. Twelve representative mistakes with commentary:

1. **شکریہ جویریہ** 😉😉 — gold 1, predicted 0 (p=0.03). Context dependence. "Thanks Javeria 😉😉" is sarcastic only relative to the tweet it answers; the post itself is a sincere thank-you with a sincere emoji (😉 = مسکراہٹ, LOVE, +). No model that sees one post in isolation can recover this; the emoji meaning actually argues for the literal reading.

2. **ٹھیک ہو جاو ناں منی اللہ پاک صحت زندگی دے آپ کو** 💙 — gold 1, predicted 0 (p=0.03). Punjabi + prayer + heart. A get-well wish with 💙 (محبت / دل). Either the label is wrong or the mockery lives in the thread; the model correctly treats prayer + heart as the sincere register.

3. **واده وکہ** 😂 and **سہولت کار** 😂 — gold 1, predicted 0 (p=0.04). Very short, irony-cue emoji. Two or three words plus 😂 (آنسو کے ہنسی, irony-cue). In the training data 😂 is only 45 % sarcastic, so the emoji alone cannot carry the decision; the words are a nickname/in-joke (واده وکہ) and a political label (سہولت کار "facilitator") whose mockery is cultural knowledge, not surface form.

4. **اتنی اردو اللہ معاف کرے** 😄 — gold 1, predicted 0 (p=0.03). Idiom + irony-cue emoji. "So much Urdu, God forgive" — the idiom کرے معاف اللہ used as a tease, with 😄 (دانت نکال کر ہنسی, cue). Our hand-crafted features flag both the idiom and the cue emoji, but the LR weight of 😄 is diluted by its many literal uses; a negative-sounding phrase with a laughing emoji is the polarity-contrast pattern that the models under-weight when the text is short.

5. **چہرہ صاف دل میں داغ منہ پہ آپ پیچھے سانپ** 🥀🔥 — gold 1, predicted 0 (p=0.04). Shayari register, mixed-polarity emojis. A couplet about two-faced people with 🥀 (مرجھایا پھول, SAD −) and 🔥 (آگ / زبردست, PRAISE +). The poetic form and the wilted flower pull strongly towards the non-sarcastic (poetry) class, which dominates 🥀 in the data (P(sarcastic| 🥀)=0.11). The sarcasm here is verbal irony inside poetry — a genre our emoji prior cannot separate.

6. **کہ میں مانتا ہوں میں برا ہوں ♥ میں فقط اتنا سچا ہوں** 🤒💪 — gold 1, predicted 0. Rare emojis. 🤒 (new moon face = شرارت, cue) is the only ironic marker and occurs 52 times in the corpus; ♥ and 💪 are glossed but neutral. Shows why the meaning/category representation matters: without the curated entry 🤒 would be an unknown byte sequence.

7. **@fawadchaudhry گھر جانے کی تیاری کرو شاباش ... ایک دوسرے پر تنقید بند کرو** — gold 0, predicted 1 (p=0.97). Ironic-praise marker, no emoji. "Stop criticising each other, pack your bags and go home, well done" — شاباش is the classic ironic closer and the sentence is a taunt; we judge the **gold label wrong** (ambiguous annotation). Both models agree with us.

8. **آپ لگے رہیں ہر کسی کی اپنی رائے ویسے میں بھی مانیکا خاندان پر تنقید کرتی ہوں** 😅😅 — gold 0, predicted 1 (p=0.97). Irony-cue emoji + "keep at it". لگے رہیں ("carry on") with 😅 (شرمندہ ہنسی, 59 % sarcastic in training) — a textbook mild-sarcasm cue; again a plausible label error.

9. **ٹٹے اٹھا رہی ہے جو اسکے پاس ہیں ہی نہیں ... بلاول نے ن لیگ سے** 😂😂😂😂😂 — gold 0, predicted 1 (p=0.96). Political mockery, negation, repeated laughter. Five 😂 (زور کی ہنسی) after a crude political joke with negation; the post is sarcastic by any reading. The dataset contains many such mocking political posts labelled 0 — the "political context" tag has a 20 % error rate for both models.

10. **@Shoaib_Jatt تنقید طریقے کی کرتا تو یوں کومنٹس میں گالیاں نہیں کھا رہا ہوتا** and five other confident false positives all contain the word **تنقید** ("criticism") and no emoji. Collection artefact. In the training split 402 posts contain تنقید and 80 % of them are sarcastic — the word seems to have been a search keyword for the sarcastic class — so both models fire on it (XLM-R predicts "sarcastic" for 96 % of test posts containing تنقید, accuracy 0.74 on them). The literal uses of the word (sincere criticism, requests to stop criticising) are then systematically misclassified. Likewise the harvesting hashtags #طنز / #مزاح / #sarcasm survive in 842 training posts (835 sarcastic) and inflate every score by ≈0.6 pt (CV ablation).

11. **@Marriyum_A #GeoNews مریم اورنگزیب کی عمران خان پر تنقید "عوام عمران خان کی تمام سازشیں جان چکی ہے" https://...** — gold 0, predicted 1 (p=0.97). Quoted speech + news headline + code-mixing. A neutral news post quoting a politician; the quote marks, the political names and تنقید all pattern-match sarcasm. Headline register with no emoji is a blind spot: 2.4 % of posts use `` … '' quotes and they are 15–19 % misclassified.

12. **حکومت کی تعریف ARY پر دیکھیں GEO حکومت پر تنقید / پر دیکھیں TV ... / بند کریں اور گھر سے باہر نکل کر دیکھیں** — gold 0, predicted 1 (p=0.97). Code-mixing, parallel structure, spelling variant (جائ). A satirical contrast between TV channels ending in the idiom دودھ کا دودھ پانی کا پانی; it is arguably sarcastic (label noise), and the Latin channel names plus the parallel "… پر دیکھیں" structure are exactly what the char n-grams learn as mockery.

Patterns. (a) Rhetorical questions (29 % error), Punjabi (24 %), negation (21 %) and spelling variants (20 %) are the hardest linguistic phenomena for both models; very short posts are surprisingly easy (5 %) because they are mostly literal replies. (b) Emoji meaning helps when the words are neutral and the emoji is a cue (😅 😳 😏 🤒), and misleads when an ironic post borrows the sincere register (😊, 💙, 🥀) — the polarity-contrast feature captures the former, nothing in the post captures the latter without thread context. (c) A large share of confident errors are annotation problems: conflicting duplicates were removed, but single-labelled posts with the opposite reading remain, and the keyword artefacts (تنقید, #طنز) make the models over-predict sarcasm on literal criticism. A cleaned re-annotation and conversational context (parent tweet) are the most promising next steps; model capacity is not the bottleneck, which is why XLM-R, the BiLSTM and TF-IDF-LR land within two points of each other.

7. Conclusion

The deployed system, selected on validation, is the greedy-selected (muril-base-cased_keep + textcnn_hybrid + xlm-roberta-base_augment) ensemble: test Macro-F1 0.824 / accuracy 83.0%; the validation-weighted ensemble reaches 0.827 / 83.3%, and the best single encoder (MuRIL, pre-trained on Urdu and Roman-Urdu) 0.813 / 81.7 %. All single-model families fall in a narrow band because the ceiling is set by label noise and context dependence rather than model capacity; only ensembling moves it. Emoji meaning matters most for short, lexically neutral posts and as a generalising representation (category/polarity) for rare emojis; hearts/flowers are the strongest non-sarcasm cue, 😅 🤣 😉 😄 lean sarcastic. Urdu code-point normalisation is a cheap consistent gain, aggressive punctuation/laughter normalisation is not. Hand-crafted phrase features (ironic praise, idioms, polarity contrast) reach 0.75 Macro-F1 alone and remain useful on top of n-grams. Future work: conversational context (parent tweet), re-annotation of conflicting posts, and Urdu-specific encoders (MuRIL) with longer training.

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